Economics S5040-S01: Dynamic Stochastic Modeling

Fall Semester 2026

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Course Description

This course develops the tools economists use to analyze decisions and equilibrium when time and uncertainty matter. We begin with intertemporal choice, production, competitive equilibrium, expected utility, and introductory asset pricing. We then study Markov processes and dynamic programming, with applications to job search, economic growth, optimal saving, and the income fluctuation problem. The second half of the course develops asset pricing in Markov environments, complete and incomplete markets, and the Aiyagari model. We conclude with a quantitative model of COVID-19 that serves as preparation for the final project.

The emphasis is on understanding the economics of dynamic models and using computation to make those models concrete. Students will formulate problems mathematically, solve them analytically when possible, implement numerical solutions in Python, and interpret the resulting policy functions, distributions, prices, and equilibria.

Readings and Software

Readings are drawn primarily from the QuantEcon lecture notes

https://quantecon.org

supplemented by course lectures. Students must have access to a computer capable of running Python and Jupyter notebooks. Computational assignments will be submitted as Jupyter notebooks through Canvas.

Learning Objectives

- Formulate intertemporal economic problems in terms of preferences, technologies, constraints, and equilibrium conditions.
- Use expected utility and measures of risk aversion to analyze choice under uncertainty.
- Construct and analyze finite-state Markov chains and stationary distributions.
- Formulate Bellman equations and solve dynamic programs using value function iteration and policy iteration.
- Apply dynamic programming to job search, saving, and income-risk problems.
- Understand the Solow growth model and the role of capital accumulation, steady states, and factor prices.
- Price risk-free assets and Lucas trees in finite-state Markov environments.
- Explain the economic distinction between complete markets, incomplete markets, and self-insurance.
- Understand how household saving decisions, wealth distributions, production, and factor prices interact in the Aiyagari model.
- Use Python to solve, simulate, visualize, and interpret dynamic economic models.

Assessment

Problem Sets and Participation: 50%
Midterm Examination: 20%
Final Project: 30%

Problem sets will combine analytical derivations, Python computation, figures, numerical verification, and economic explanation. Students may consult course materials, classmates, and AI tools, but are responsible for understanding and being able to explain every result they submit.

Course Schedule

Week 1 - Intertemporal Choice and General Equilibrium

Preferences and utility; dated commodities; intertemporal budget constraints; production possibilities; the planner's problem; competitive equilibrium; an introduction to the First Welfare Theorem.

Reading: Course Lectures 1-2.

https://intro.quantecon.org/pv.html

https://intro.quantecon.org/cons_smooth.html

Week 2 - Expected Utility and Introductory Asset Pricing

von Neumann-Morgenstern expected utility; absolute and relative risk aversion; state-contingent claims; state prices; stochastic discount factors.

Reading: Course Lectures 3-4.

https://intro.quantecon.org/monte_carlo.html

Week 3 - Markov Chains

Finite-state Markov chains, transition matrices, multi-step probabilities, simulation, and stationary distributions.

Reading: QuantEcon: Finite Markov Chains.

https://intro.quantecon.org/markov_chains_I.html

https://intro.quantecon.org/markov_chains_II.html

Week 4 - Dynamic Programming: Discrete Problems

Bellman equations and value functions; simple finite-state problems; value function iteration; policy iteration and Howard's method.

Reading: Course material; QuantEcon: Discrete State Dynamic Programming (supplementary).

https://python-advanced.quantecon.org/discrete_dp.html

Week 5 - The McCall Job Search Model

Job search as a dynamic programming problem; continuation values; reservation wages; a continuous state space.

Reading: QuantEcon: Job Search I - The McCall Search Model.

https://python.quantecon.org/mccall_model.htm

https://python.quantecon.org/mccall_model_with_separation.html

Week 6 - The Solow Growth Model

Production, capital accumulation, steady states, convergence, comparative statics, and factor prices. Saving is exogenous in Solow and will become a choice in Week 7.

Reading: QuantEcon: selected Solow sections from Using Newton's Method to Solve Economic Models.

https://intro.quantecon.org/solow.html

https://python.quantecon.org/newton_method.html

Week 7 - Optimal Savings

Intertemporal consumption and saving; Bellman and Euler equations; consumption and saving policy functions; numerical solution.

Reading: QuantEcon: Optimal Savings I: The Cake-Eating Problem; selected material from Optimal Savings II.

https://python.quantecon.org/os.html

https://python.quantecon.org/os_numerical.html

https://python.quantecon.org/os_stochastic.html

Week 8 - The Income Fluctuation Problem

Idiosyncratic income risk; borrowing constraints; consumption smoothing; precautionary saving; self-insurance.

Reading: QuantEcon: Income Fluctuation Problem I: Discretization and VFI.

https://python.quantecon.org/ifp_discrete.html

https://python.quantecon.org/ifp_opi.html

https://python.quantecon.org/ifp_egm.html

https://python.quantecon.org/ifp_egm_transient_shocks.html

https://python.quantecon.org/ifp_advanced.html

Week 9 - Review and Midterm

Review of Weeks 1-8 and midterm examination.

Reading: No new reading.

Week 10 - Asset Pricing in a Markov Economy

Markov consumption and dividend processes; stochastic discount factors; recursive pricing equations; state-dependent risk-free rates.

Reading: QuantEcon: Asset Pricing: Finite State Models.

https://python.quantecon.org/harrison_kreps.htm

Week 11 - The Lucas Tree and Risk-Free Rates

The Lucas tree in a finite-state Markov economy; equilibrium asset prices; risk-free claims; expected returns and economic interpretation.

Reading: Continue QuantEcon: Asset Pricing: Finite State Models.

https://python.quantecon.org/markov_asset.html

Week 12 - Complete Markets and Risk Sharing

Arrow securities; state-contingent trade; competitive equilibrium with complete markets; efficient risk sharing.

Reading: QuantEcon: selected sections of Competitive Equilibria with Arrow Securities.

https://python.quantecon.org/ge_arrow.html

Week 13 - Incomplete Markets

Limited asset trade, self-insurance, precautionary saving, and the contrast between complete and incomplete markets.

Reading: Revisit QuantEcon: Income Fluctuation Problem I; selected comparison with the Week 12 Arrow-securities economy.

https://python.quantecon.org/lq_permanent_income.html

https://python.quantecon.org/lq_bewley_complete_markets.html

Week 14 - The Aiyagari Model

Heterogeneous households with idiosyncratic income risk; borrowing constraints; stationary wealth distributions; aggregate capital; endogenous wages and interest rates; general equilibrium.

Reading: QuantEcon: The Aiyagari Model.

https://python.quantecon.org/aiyagari.html

Week 15 - Modeling COVID-19 and Final Project

A quantitative epidemic model; SIR/SEIR dynamics, transmission and mitigation; economic interpretation and preparation for the final project on the economic costs of the COVID lockdown.

Reading: QuantEcon: Modeling COVID-19.

https://python.quantecon.org/sir_model.html

Final Project

The final project will build on the QuantEcon COVID-19 model studied in Week 15. Students will use the tools developed throughout the semester to analyze a quantitative question concerning the economic costs of the COVID lockdown. Detailed project instructions will be distributed separately.